Customer Shopping Behavior Analysis

Uncovering purchasing patterns, customer segments and revenue drivers using Python, SQL and Power BI

Item	Details
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Role focus	Data Analyst / Business Analyst
Dataset	Customer Shopping Behavior
Tools	Python (pandas), SQL Server (T-SQL), Power BI,
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1. Executive Summary

This project analyses the shopping behaviour of 3,900 retail customers to understand who buys, what they buy, how they pay for delivery, and which customer groups generate the most value. The work follows an end-to-end analytics workflow: Python was used to clean and prepare the data, SQL Server was used to answer nine business questions, and Power BI was used to build an interactive dashboard for decision makers.

Headline findings

- **Revenue totals USD 233,081.** Male customers contribute 67.7% of it (USD 157,890) mainly because they are 68% of the customer base; average spend per purchase is almost identical for men and women.
- **Subscribers do not spend more.** Subscribed and non-subscribed customers both average about USD 59 per purchase, yet frequent buyers are more likely to subscribe (27.6% vs 22.4%).
- **Discounts are widespread but not lifting basket size.** 43% of purchases use a discount, and discounted orders average USD 59.28 versus USD 60.13 without a discount.
- **The customer base is loyal but not growing.** 80% of customers have more than 10 previous purchases, while only 2.1% are new.
- **Clothing leads the portfolio** with 44.7% of revenue; Young Adults (18-31) are the top-revenue age group at 26.7%.

The report closes with seven data-backed recommendations covering customer acquisition, subscription design, discount strategy and category growth.

2. Project Objectives and Business Problem

A retail business collects transaction-level data but needs clear answers to everyday commercial questions: which customers are most valuable, whether subscription and discount programmes are working, and where to focus marketing effort. The objectives of this project were to:

1. Clean and prepare raw customer data so it is reliable for analysis.
2. Engineer useful features such as age groups and purchase frequency in days.
3. Use SQL to quantify revenue, spending behaviour, product popularity and customer loyalty.
4. Build an interactive Power BI dashboard that lets non-technical users filter and explore the results.
5. Translate the numbers into actionable business recommendations.

3. Dataset Overview

The dataset is a publicly available secondary dataset from Kaggle. Each of the 3,900 rows describes one customer and the purchase they made, together with demographic, product and behavioural attributes.

Group	Columns
Customer	Customer ID, Age, Gender, Location (50 US states)
Product	Item Purchased (25 items), Category (4), Size, Color, Season
Transaction	Purchase Amount (USD), Review Rating, Payment Method, Shipping Type
Marketing / loyalty	Subscription Status, Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases

Key data facts: purchase amounts range from USD 20 to USD 100 (mean USD 59.76); review ratings range from 2.5 to 5.0 (mean 3.75); customer ages range from 18 to 70; 73% of customers are not subscribed.

4. Tools and Technologies

Tool	Purpose in this project
Python (pandas)	Data inspection, cleaning, feature engineering and loading the data into SQL Server
Jupyter Notebook	Interactive development and documentation of the cleaning steps
SQL Server (T-SQL)	Aggregations, CASE logic, CTEs and window functions for business analysis
Power BI	Interactive dashboard with KPI cards, slicers and charts

5. Data Cleaning and Preprocessing (Python)

The raw CSV was loaded into a pandas DataFrame and profiled with info(), describe() and isnull().sum(). The profile showed 3,900 rows and 18 columns with no duplicates in Customer ID and a single data-quality issue: 37 missing values in Review Rating.

5.1 Handling missing values

Rather than dropping rows or filling with one global average, missing ratings were replaced with the median rating of the same product category. This keeps all 3,900 records and respects the fact that categories can have different rating levels.

```
df["Review Rating"] = df.groupby("Category")["Review Rating"].transform(
    lambda x: x.fillna(x.median())
)
```

5.2 Standardising column names

Column names were converted to lowercase snake_case so they can be used directly in SQL and Power BI without brackets or quotes.

```
df.columns = df.columns.str.lower().str.replace(" ", "_")
df = df.rename(columns={"purchase_amount_(usd)": "purchase_amount"})
```

5.3 Feature engineering

Two analytical features were created. Age was split into four equal-sized groups using quartiles, and the text frequency labels were converted into a numeric number of days between purchases.

```
labels = ["Young Adult", "Adult", "Middle-aged", "Senior"]
df["age_group"] = pd.qcut(df["age"], q=4, labels=labels)

frequency_mapping = {"Weekly": 7, "Fortnightly": 14, "Bi-Weekly": 14,
    "Monthly": 30, "Quarterly": 90, "Every 3 Months": 90, "Annually": 365}
df["purchase_frequency_days"] = df["frequency_of_purchases"].map(frequency_mapping)
assert df["purchase_frequency_days"].notna().all()   # guard against unmapped labels
```

Age group	Age range	Customers
Young Adult	18 - 31	1,028
Adult	32 - 44	942
Middle-aged	45 - 57	986
Senior	58 - 70	944

5.4 Removing a redundant column

A comparison showed that discount_applied and promo_code_used contain identical values in all 3,900 rows. Keeping both would double-count the same information, so promo_code_used was dropped.

```
(df["discount_applied"] == df["promo_code_used"]).all()   # True
df = df.drop("promo_code_used", axis=1)
```

5.5 Loading the data into SQL Server

The cleaned DataFrame was written to a SQL Server table using SQLAlchemy and pyodbc with Windows authentication, creating a seamless hand-off between Python and SQL.

```
engine = create_engine("mssql+pyodbc:///?odbc_connect=" + quote_plus(connection_string))
df.to_sql("customer_shopping_behavior", engine, if_exists="replace", index=False)
```

Step	Before	After
Missing review ratings	37	0
Columns	18	18 (2 added, 1 removed, renamed)
Rows	3,900	3,900 (no data lost)

6. SQL Analysis

Nine business questions were answered in SQL Server against the table customer_shopping_behavior. The queries use aggregation, conditional logic with CASE, common table expressions (CTEs) and window functions. Results below were verified against the source data.

Q1. What is the total revenue generated by male vs female customers?

```
SELECT gender, SUM(purchase_amount) AS revenue
FROM customer_shopping_behavior
GROUP BY gender;
```

Gender	Revenue (USD)
Male	157,890

Gender	Revenue (USD)
Female	75,191

Insight: Men generate 67.7% of revenue, but average spend is nearly equal (USD 59.54 male vs USD 60.25 female). The gap comes from customer count (2,652 vs 1,248), not spending power.

Q2. Which are the top 5 products with the highest average review rating?

```
SELECT TOP 5 item_purchased,
       ROUND(AVG(CAST(review_rating AS DECIMAL(10,2))), 2) AS average_product_rating
FROM customer_shopping_behavior
GROUP BY item_purchased
ORDER BY AVG(CAST(review_rating AS DECIMAL(10,2))) DESC;
```

Rank	Product	Average rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Insight: Ratings are tightly clustered (3.78 to 3.86, against an overall mean of 3.75), so no product is a clear quality outlier.

Q3. How does average purchase amount compare between Standard and Express shipping?

```
SELECT shipping_type, ROUND(AVG(purchase_amount), 2) AS avg_purchase_amount
FROM customer_shopping_behavior
WHERE shipping_type IN ('Standard', 'Express')
GROUP BY shipping_type;
```

Shipping type	Avg purchase (USD)
Standard	58.46
Express	60.48

Insight: Express customers spend about USD 2 more (+3.4%) per purchase, a modest difference.

Q4. Do subscribed customers spend more than non-subscribed customers?

```
SELECT subscription_status, COUNT(customer_id) AS total_customers,
       ROUND(AVG(purchase_amount), 2) AS avg_spend,
       SUM(purchase_amount) AS total_revenue
FROM customer_shopping_behavior
GROUP BY subscription_status;
```

Subscribed	Customers	Avg spend (USD)	Total revenue (USD)
Yes	1,053	59.49	62,645
No	2,847	59.87	170,436

Insight: No. Subscribers spend the same as non-subscribers (USD 59.49 vs USD 59.87). Only 27% of customers subscribe, so the programme currently adds little measurable spend.

Q5. Which five products have the highest percentage of discounted purchases?

```
SELECT TOP 5 item_purchased,
       ROUND(100.0 * SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END)
             / COUNT(*), 2) AS discount_rate
FROM customer_shopping_behavior
GROUP BY item_purchased
ORDER BY discount_rate DESC;
```

Product	Discount rate (%)
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

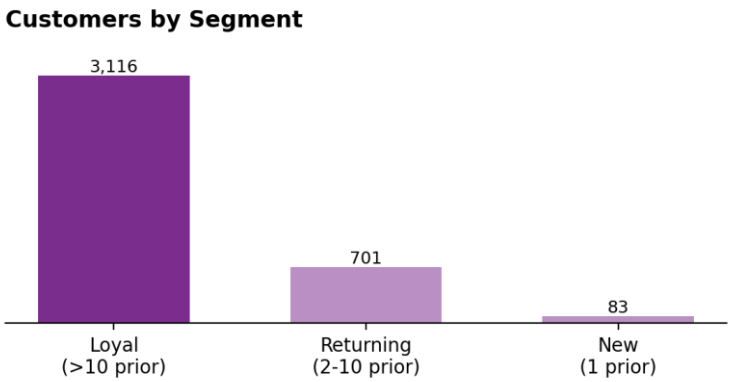
Insight: Roughly half of all purchases of these products are discounted. Because discounted orders average USD 59.28 versus USD 60.13 at full price, discounts are not increasing basket value.

Q6. How can customers be segmented into New, Returning and Loyal groups?

Segments were defined on previous purchases: New = 1, Returning = 2 to 10, Loyal = more than 10.

```
WITH customer_type AS (
  SELECT customer_id,
    CASE WHEN previous_purchases = 1 THEN 'New'
        WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
        ELSE 'Loyal' END AS customer_segment
  FROM customer_shopping_behavior)
SELECT customer_segment, COUNT(*) AS number_of_customers
FROM customer_type GROUP BY customer_segment;
```

Segment	Customers	Share
Loyal	3,116	79.9%
Returning	701	18.0%
New	83	2.1%



> **Insight:** The business relies on an established, repeat-buying base. Very few new customers (83) indicates a weak acquisition pipeline.

Q7. What are the top three most purchased products within each category?

```
WITH item_counts AS (
 SELECT category, item_purchased, COUNT(customer_id) AS total_orders,
      ROW_NUMBER() OVER (PARTITION BY category
                  ORDER BY COUNT(customer_id) DESC) AS item_rank
 FROM customer_shopping_behavior
 GROUP BY category, item_purchased)
SELECT item_rank, category, item_purchased, total_orders
FROM item_counts WHERE item_rank <= 3;
```

Category	#1	#2	#3
Accessories	Jewelry (171)	Belt (161)	Sunglasses (161)
Clothing	Blouse (171)	Pants (171)	Shirt (169)
Footwear	Sandals (160)	Shoes (150)	Sneakers (145)
Outerwear	Jacket (163)	Coat (161)	-

> **Insight:** Demand is evenly spread: leaders differ from runners-up by only a few orders. Outerwear has just two products, which is why it shows only two ranks.

Q8. Are customers with more than five previous purchases more likely to subscribe?

```
SELECT subscription_status, COUNT(customer_id) AS repeat_buyers
FROM customer_shopping_behavior
WHERE previous_purchases > 5
GROUP BY subscription_status;
```

Subscribed	Repeat buyers (>5 purchases)
Yes	958
No	2,518

Counts alone cannot show likelihood, so the subscription rate was compared across both groups:

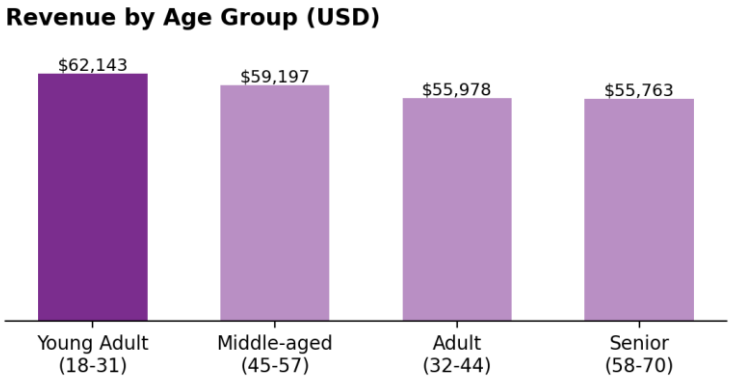
Previous purchases	Customers	Subscription rate
5 or fewer	424	22.4%

Previous purchases	Customers	Subscription rate
More than 5	3,476	27.6%

> **Insight:** Yes, modestly. Repeat buyers subscribe about 5 percentage points more often. The 2,518 repeat buyers who have not subscribed are the most natural target for a conversion campaign.

Q9. Which age group contributes the most to total revenue?

```
SELECT age_group, SUM(purchase_amount) AS total_revenue
FROM customer_shopping_behavior
GROUP BY age_group
ORDER BY total_revenue DESC;
```



> **Insight:** Young Adults (18-31) lead with USD 62,143 (26.7%), but the spread to Seniors (23.9%) is small, so revenue is balanced across all ages.

7. Key Business Insights

Area	Evidence	What it means
Gender	Men 67.7% of revenue; avg basket USD 59.54 vs USD 60.25	Revenue gap is a reach gap, not a spending gap
Subscription	27% subscribed; avg spend USD 59.49 vs USD 59.87	Programme has not lifted spend
Loyalty	80% Loyal, 18% Returning, 2.1% New	Strong retention, weak acquisition
Discounts	43% of orders discounted; USD 59.28 vs USD 60.13	Discounts may be eroding margin without raising basket size
Shipping	Express USD 60.48 vs Standard USD 58.46	Small uplift of about 3.4%
Category	Clothing 44.7%, Accessories 31.8%, Footwear 15.5%, Outerwear 7.9% of revenue	Revenue concentrated in Clothing and Accessories
Age	Young Adults 26.7%, Seniors 23.9%	Balanced appeal across all age groups

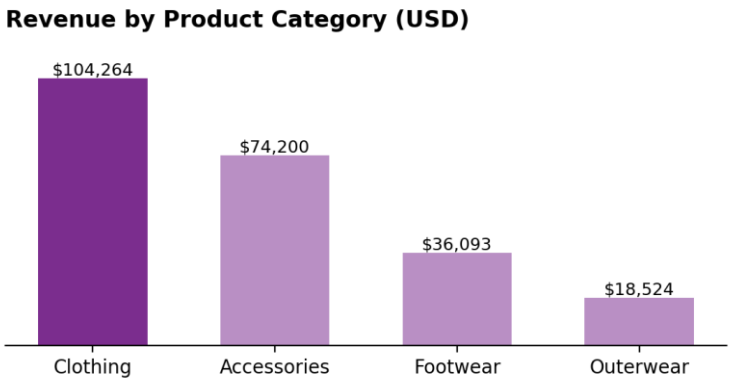


Figure 1: Clothing and Accessories together account for 76.5% of revenue.

8. Dashboard Development (Power BI)

The cleaned data was imported into Power BI to create a one-page interactive dashboard so that managers can explore results without writing SQL.

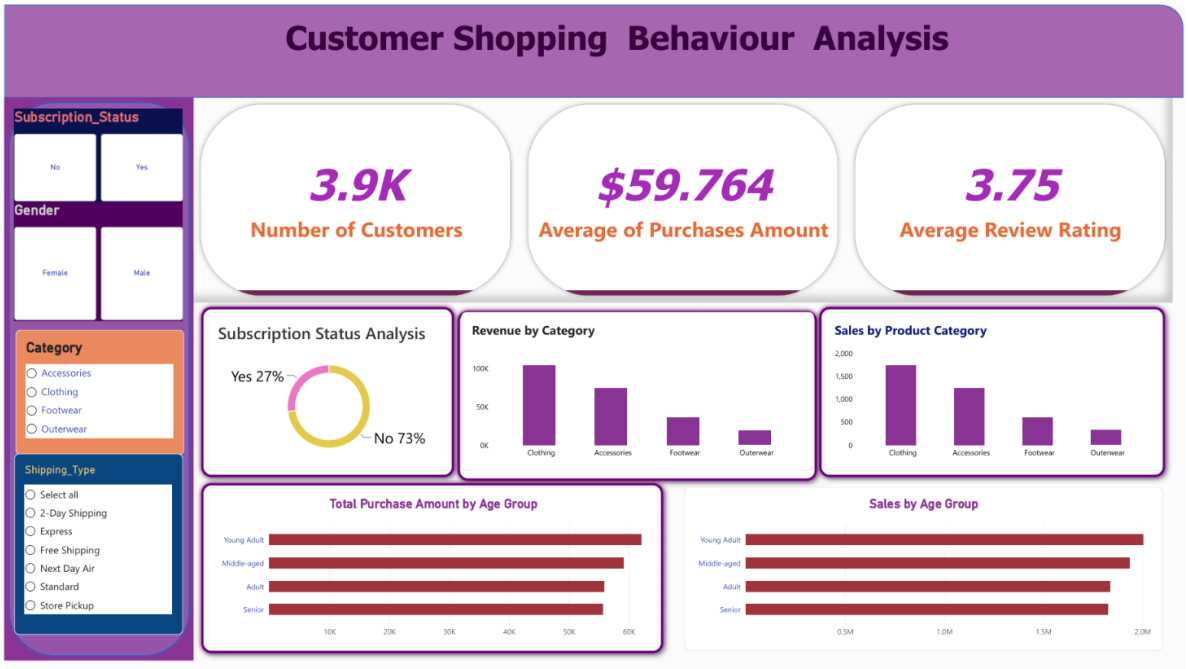


Figure 2: Customer Shopping Behaviour Analysis dashboard (Power BI).

8.1 Dashboard components

- **KPI cards:** 3.9K customers, average purchase amount of about USD 59.76 and average review rating of 3.75, giving an instant health check.
- **Subscription Status Analysis:** donut chart showing 73% non-subscribers and 27% subscribers.
- **Revenue by Category and Sales by Product Category:** column charts showing Clothing as the leading category by both revenue and number of orders.
- **Total Purchase Amount by Age Group:** horizontal bars confirming Young Adults as the top-revenue segment.

- **Slicers:** Subscription Status, Gender, Category and Shipping Type filters update every visual at once.

8.2 Design choices

A consistent purple palette with high-contrast headings was used for readability. KPIs sit at the top, filters on the left and detailed charts below, following a top-to-bottom flow from summary to detail. Every chart answers one question, avoiding clutter.

9. Recommendations

1. **Grow the female customer base.** Women are only 32% of customers but spend as much per order as men. Targeted campaigns and product ranges could raise revenue without needing higher basket values.
2. **Redesign the subscription programme.** Add tangible benefits (free shipping, early access, members-only items) so that subscribing leads to higher spend, and test the impact.
3. **Convert repeat buyers into subscribers.** Target the 2,518 non-subscribed customers with more than five purchases; they already show loyalty and subscribe at a higher rate.
4. **Make discounting more selective.** Discounts apply to 43% of orders without raising spend. Run A/B tests that reduce blanket discounts on Hat, Sneakers, Coat, Sweater and Pants and measure the effect on volume and margin.
5. **Invest in new-customer acquisition.** With only 2.1% new customers, introduce welcome offers and referral rewards, while protecting the loyal base with a points or tier programme.
6. **Use Express shipping as an upsell, not a growth lever.** The spend uplift is small (about USD 2), so promote it at checkout but avoid heavy investment.
7. **Expand and promote Outerwear.** It is the weakest category (7.9% of revenue, only two products, lowest average spend of USD 57.17). Widening the range and running seasonal campaigns could unlock growth.

10. Limitations and Future Work

Limitations. The dataset has no date field, so trends over time cannot be measured. Each row represents one purchase per customer, so "customers" and "purchases" are treated as equivalent. Differences between groups are often small, and no statistical significance tests were run. Public datasets may also be simulated, so findings illustrate analytical method rather than guaranteed real-world behaviour.

Future work. Add hypothesis tests (t-tests, chi-square) to confirm which differences are significant; build an RFM (Recency, Frequency, Monetary) model; use DAX measures for dynamic KPIs; and train a simple classification model to predict subscription likelihood.

11. Conclusion

This project demonstrates a complete analytics workflow: cleaning and engineering data in Python, answering nine business questions in SQL, and communicating results through an interactive Power BI dashboard. The analysis shows a loyal but narrow customer base, a subscription and discount strategy that is not clearly raising spend, and clear opportunities in female customers, repeat-buyer conversion and the Outerwear category. The recommendations convert these findings into practical next steps for a business team.

Skills Demonstrated

Skill area	How it was applied
Data cleaning (Python / pandas)	Profiled data, imputed 37 missing ratings by category median, standardised names, removed a redundant column
Feature engineering	Quartile-based age groups, numeric purchase-frequency feature, customer loyalty segments
SQL	Aggregations, CASE logic, CTEs, window functions (ROW_NUMBER), percentage calculations
Data visualisation (Power BI)	KPI cards, donut and bar charts, cross-filtering slicers, consistent design
Business analysis	Converted findings into seven prioritised, evidence-based recommendations
Data validation	Checked query results against independent calculations and added assertions in code